

Current Scenario

Product images come from multiple vendors

No control over:

- Aspect ratios
- Backgrounds
- Framing
- Positioning

Impact on Business

- Product visibility issues (cropped or too small)
- Inconsistent UI → poor user experience
- Misaligned thumbnails → lower trust
- Reduced conversion rates



Low Product Coverage



Image not fitting in standardized container ratio, leading to white space BG



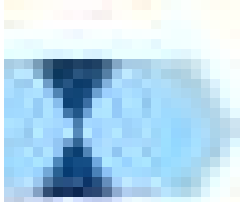
Cropped Product thumbnails



Model Crop



Product / Model not at centre



Sliikk Aspect Ratio Fix



Proposed Solution Workflow: Product & BG Differentiation + BG Crop + BG Stretch

We use a two-stage segmentation pipeline: a domain-specific transformer-based model followed by a general-purpose Foundation Segmentation model, both producing pixel-level masks which are converted into bounding regions

Product (G)

Stage	Condition (Trigger)	What Happens	Model / Method Used	Output / Action	Stop Condition	Done
Step 0: Aspect Ratio Identification	Image received	Extract aspect ratio from: - URL (preferred) - Image dimensions (fallback)	Regex + OpenCV	Calculate W/H ratio	Ratio > 0.75 -> STOP	
Step 1: Product Boundary detection + crop	Ratio > 0.75	Detect product + background	Flagship Face Model (Face-fusion-parser)	Generate segmentation mask + bounding box + attempt safe crop	1. Crop extent until image reaches target aspect ratio OR 2. Until Product boundary saturation	BACKGROUND SUBTRACTION / CROP
Step 2 (Fallback) HF model fails to produce a 'sufficiently' valid bounding region	Product not in trained/demo of HF model & ratio < 0.75	Detect product using general segmentation	SAM (Segment Anything - Meta)	Generate masks + filter + union + bounding box + safe crop	Same as above	BACKGROUND SUBTRACTION / CROP



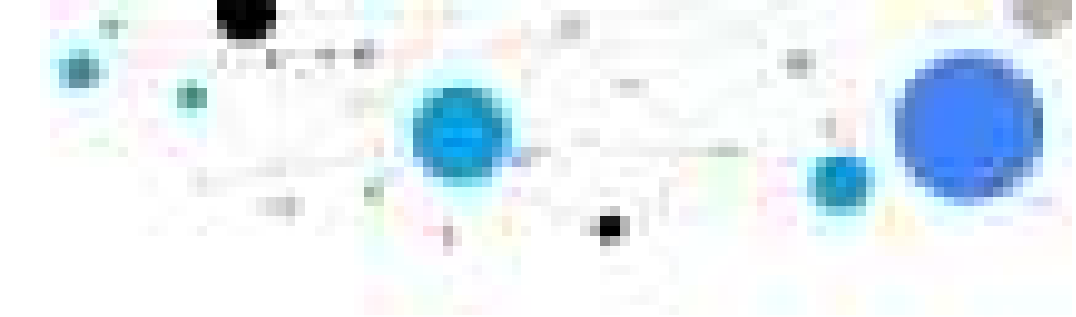
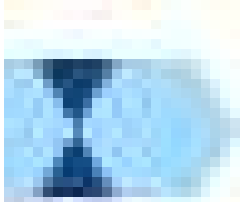
STEP 1
BORDER DETECTION
&
BG CROP



STEP 2
NA

STEP 3
BG STRETCH





Sliikk Aspect Ratio Fix

Proposed Solution Workflow: Product & BG Differentiation + BG Crop + BG Stretch

Stage	Condition (Trigger)	What Happens	Model / Method Used	Output / Action	Stop Condition	Base
Step 2 - BG STRETCH	Ratio ≥ 0.75	Adjust background only	Variance-based BG classification + OpenCV resize	Stretch background (horizontal / vertical)	Limit Ratio = 0.75 for solid/gradient image (Low val) Limited stretch for complex BGs (25% of BG W/H)	BACKGROUND ADDITION / PADDING
Final Step	After all transformations	Separate Product Box at end	Plan: monitoring of product bounding box not executed	Final standardised image	END	IMAGE POSITIONING



STEP 1
BORDER DETECTION
&
BG CROP



STEP 2
NA



STEP 3
BG STRETCH



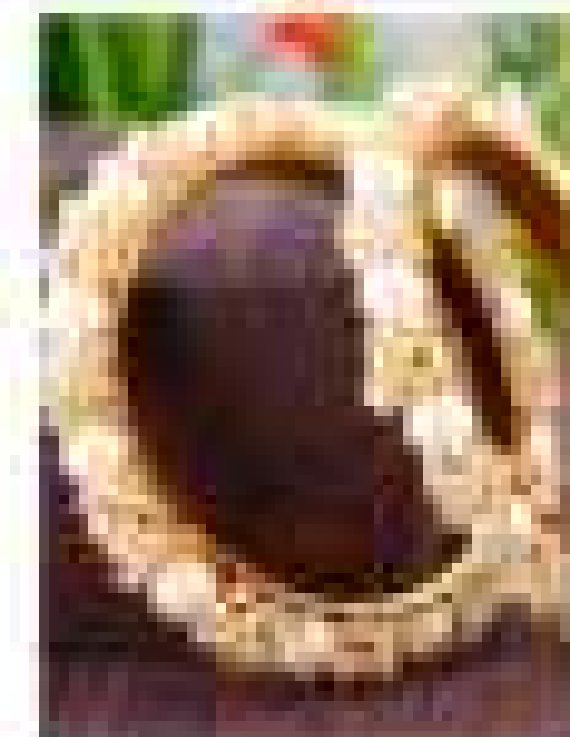
Hard Constraints:

- Product mask (Object Boundary) treated as immutable region
- All transformations applied only on background pixels

Sliikk Aspect Ratio Fix

MVP Code Flaws

1. Float Error: Aspect Ratio reaches near 0.75 (eg: 0.7495/0.7505) and pipeline stops
2. SAM Model not working - which was designed as backup for product Fashn Parser not trained
3. BG Stretch - Replicating and stretching replicated BG in some cases. Expected: stretching pixel across BG, no BG replication. Works for solid, gradient, textured BG only



Model not trained to identify Bangles



Model recognized all image as BG, and stretched all image to fit from 1 to 0.75



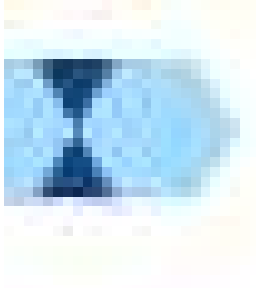
Img ratio = 1

Wrong Detection saved by:

Crop Condition: 1: Crop extent until image reaches target aspect ratio OR 2: Until Product boundary saturates
GDAL: Crop image to achieve aspect ratio = 0.75, not achieve maximum product coverage

Additional Conditions required to improve in MVP:

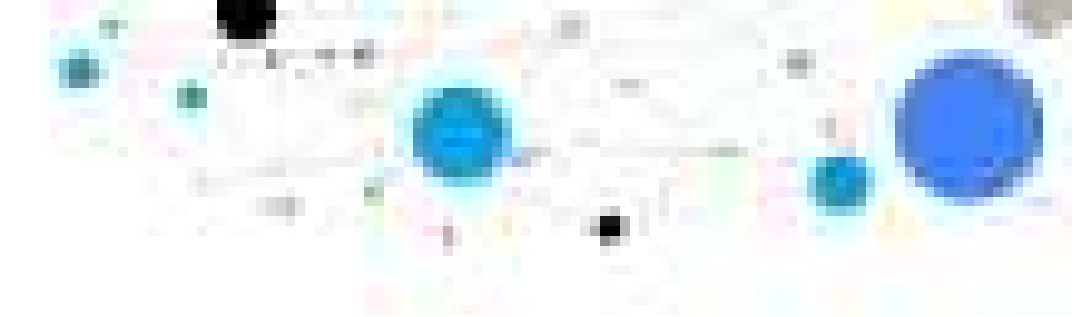
1. 1% Gap b/w Product Border and Container to avoid product edge touching container, that feels like cropped product
2. Signature, Text, Tags inclusion in product boundary
3. Recenter product if bg available



Slikk Aspect Ratio Fix

MVP Model Fixes ~87% Images

Analysis 



Working Pipeline Examples

Original

Step1 (RED bbox)

0.5215

0.5568



0.5568

0.7497



0.5617

0.7280



0.7280

0.7500



1.6000

0.8700



0.8700

0.7500



0.8277

0.7501

